Part: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3111**

Title: **STRESS IN CURVED BEAMS SECTIONS**

Revision: **1.0**

Date: **23/09/2022**

1 INTRODUCTION

1.1 OBJECT, SCOPE AND LIMITATIONS

The purpose of this chapter is to propose a correction of beam stresses presented in chapter §3110 "BEAMS SECTIONS", when the beam is not straight. In addition to applicable limitations presented in that chapter for slender beams, the simplifications presented here are valid for slightly curved beams, i.e. with a ratio between curvature radius *R* and beam section height *h* greater than 8, loaded in its plane of curvature (which is the case typical of fuselage frames).

2 METHOD DESCRIPTION

The correction of beam stresses, when beam is not straight, may be performed according to the formulation defined in Ref. 1, chapter 9.

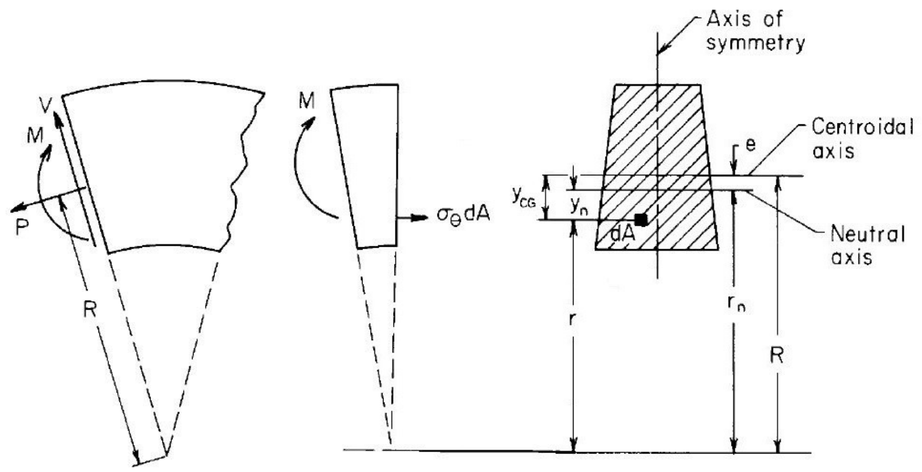

Figure 1. Curved beam section in bending, according to figure 9.1 of Ref. 1

Longitudinal (circumferential) stress in pure bending can be obtained as follows…

$\sigma_\vartheta = M \cdot y_n / (A \cdot e \cdot r)$ …where: $e = I / (R \cdot A)$

$\sigma_\vartheta = M \cdot y_n / (A \cdot e \cdot r) = (M \cdot y_n / I) \cdot R / r = (M \cdot y_n / I) \cdot R / (R - y_{cg}) = (M \cdot y_n / I) \cdot 1 / (1 - y_{cg}/R)$

… thus, equal to the bending stress calculated for the section as it were part of a straight beam, multiplied by the factor $1/(1-y_{cg}/R)$, with R the curvature radius of the centroidal axis and y_{cg} the coordinate of the point in the local reference of the section with origin in its centre of gravity (note the sign convention for this coordinate is to be positive when the point is between the centre of gravity and the curvature axis of the beam).

The shear stress resulting for above mentioned bending stress distribution, at a point of the section at a distance "r" from the curvature center, is:

$$\tau_{r\theta} = \frac{V(R-e)}{t_r A e r^2}(RA_r - Q_r)$$ …where: $$A_r = \int_b^r dA_1 \quad \text{and} \quad Q_r = \int_b^r r_1\, dA_1$$

The radial stress induced by bending within the curved beam is (see details in Ref. 1 , sub-chapter 9.1):

$$\sigma_r = \frac{R-e}{t_r A e r} M \left(\int_b^r \frac{dA_1}{r_1} - \frac{A_r}{R-e} \right)$$

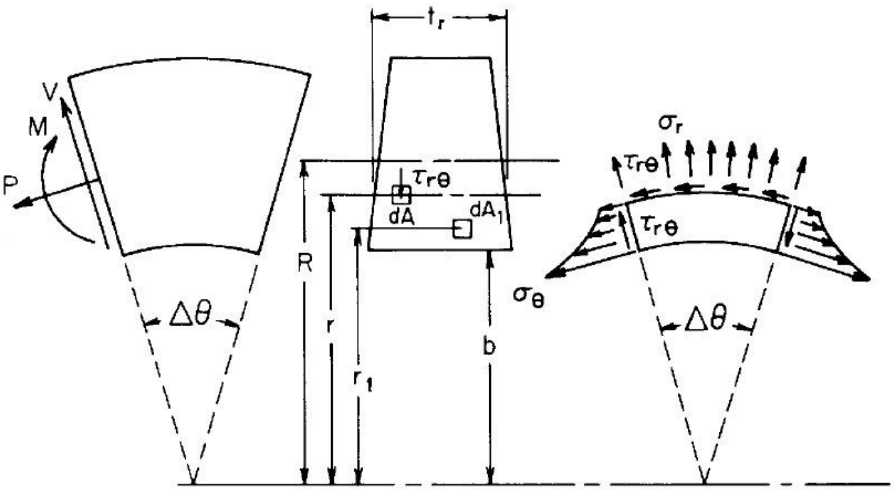

Figure 2. Stress at inner point in section of a curved beam in bending, acc. to figure 9.2 of Ref. 1

The assumption of uniform stress distribution obtained for pure axial load at the centre of gravity of the section is considered still valid for slightly curved beams.

Additional remarks:

- ✓ For curved beams with ratios of curvature radius to section height greater than 50 these curvature corrections shall not be necessary, they will be treated as straight beams.

- ✓ When internal loads are obtained from FEM analysis and the beam model include elements to capture curvature effects (in hoop, shear and radial stresses), additional curvature corrections shall neither be necessary.

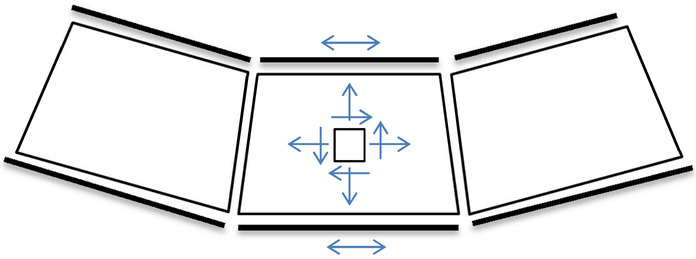

Figure 3. Curved beam in GFEM including 2D elements for the web segment

- ✓ An acceptable approximation of the radial stress in the web of flanged beams in bending, where bending is principally supported by differential axial forces in outer and inner flanges, is the following:

$$\sigma_r = P_i / (r \cdot t_w)$$

… with P_i the axial load at the inner flange, t_w is the thickness of the web and r the curvature radius at the analysis point. Note P_i and σ_r have consistent sign convention in that formula: positive tension load leads to positive tension radial stress.

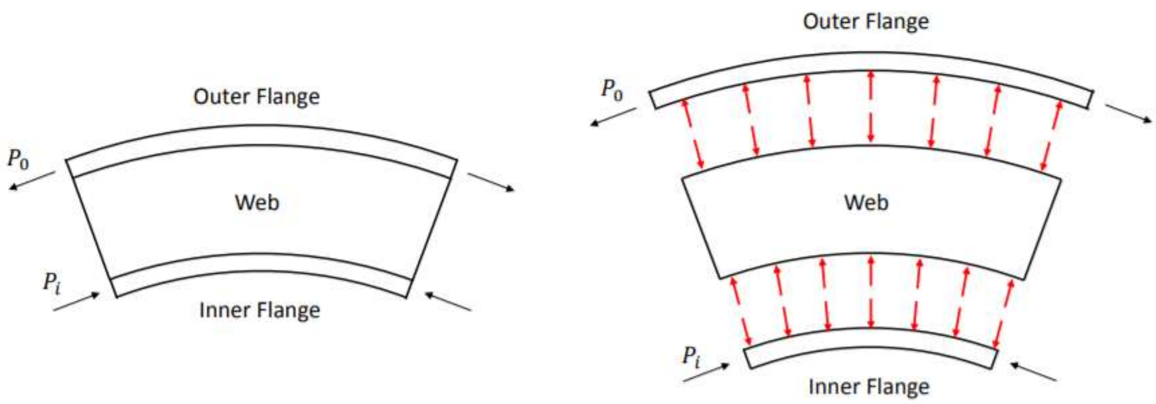

Figure 4. Radial stress in flanged beams

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
A	mm^2	Area of the beam cross-section
e	mm	Distance from neutral axis to section centroid
h	mm	Height of the beam cross-section
I	mm^4	Moment of inertia, or second moment of area, of the cross-section at its centroid
i	-	As subscript, relative to frame inner flange
M	N·mm	Bending moment
o	-	As subscript, relative to frame outer flange
P	N	Axial force
Q	mm^3	Static moment, or first moment of area, of the cross-section, ref. to curvature axis
R	mm	Radius of curvature of cross-section's centroid
r	mm / -	Radial distance or, as subscript, relative to radial directions
t	mm	Thickness of the beam cross-section
TBC		To Be Confirmed
V	N	Shear force
w	-	As subscript, relative to frame web
y_{cg}	mm	Location of point in section, ref. to centroid axis
y_n	mm	Location of point in section, ref. to neutral axis
σ	MPa	Axial/bending stress
τ	MPa	Shear stress
θ	-	As subscript, relative to circumferential direction

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[W. C. Young & R. G. Budynas]	7th Ed.	Roark's Formulas for Stress and Strain

SOFTWARE

Program	Version	Description
-	-	-
-	-	-

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 19/09/2022	G. Baños & G.Moreda	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 19/09/2022	Florian Glock 23/09/2022

Table 3. *Approval signatures table for last revision of this chapter*